Sequence & Series Practice

Nine Education IIT Academy | Class 11 Mathematics | 100 Problems: AP, GP, HP, AGP & Square/Cube Series

Questions

1. The 7th term of an AP is 34 and the 13th term is 64. Find the 20th term.

2. If the sum of first n terms of an AP is $S_n = 3n^2 + 5n$, find its 10th term.

3. Insert 3 arithmetic means between 2 and 18.

4. Find the sum of first 15 terms of an AP with first term 3 and common difference 4.

5. Which term of the AP 3, 8, 13, ... is 78?

6. Find the sum of all natural numbers between 1 and 100 divisible by 5.

7. If 2k+1, 3k+5, 5k+11 are in AP, find k.

8. The sum of three numbers in AP is 27 and their product is 504. Find the numbers.

9. Find the sum of first 20 terms of an AP whose nth term is 3n+2.

10. Is 301 a term of the AP 5, 11, 17, ...?

11. Find the sum of first 25 terms of the AP 7, 10, 13, ...

12. The angles of a triangle are in AP, and the greatest is double the smallest. Find the angles.

13. Find the sum of all 3-digit numbers divisible by 7.

14. If the pth term of an AP is 1/q and the qth term is 1/p, find the sum of the first pq terms.

15. How many terms of the AP 9, 17, 25, ... must be taken to give a sum of 636?

16. Find three numbers in AP whose sum is 15 and product is 105.

17. The sums of n terms of two APs are in the ratio (3n+8):(7n+15). Find the ratio of their 12th terms.

18. Find the middle term of the AP 1, 8, 15, ..., 505.

19. If a, 7, b, 23, c are in AP, find a, b, c.

20. Find the sum of the series 2+5+8+...+(3n−1) in terms of n.

21. In a GP, the 4th term is 24 and the 7th term is 192. Find the first term and common ratio.

22. Find the sum of first 6 terms of the GP 2, 6, 18, ...

23. Find the sum to infinity of the GP 8, 4, 2, 1, ...

24. Insert two geometric means between 4 and 32.

25. Find the nth term of the GP 5, 10, 20, ...

26. The sum to infinity of a GP is 15 and the sum of squares of its terms is 45. Find the first term.

27. Three numbers with sum 21 are in AP; if 2, 3, 9 are added respectively they form a GP. Find the numbers.

28. Find the sum of first 5 terms of a GP with a=1, r=1/3.

29. The 5th term of a GP is 81 and the 2nd term is 3. Find the first term and common ratio.

30. A ball dropped from 100 m rebounds to 3/4 of its height each time. Find the total distance travelled before it comes to rest.

31. Find the 6th term of a GP whose first term is 3 and common ratio is −2.

32. The sum of first three terms of a GP is 13/3 and their product is 1. Find the terms.

33. If the 1st and nth terms of a GP are a and b, and P is the product of the first n terms, show that $P^2=(ab)^n$.

34. Find the sum to infinity of the GP 1 − 1/3 + 1/9 − 1/27 + ...

35. The first term of a GP is 1. The sum of the 3rd and 5th terms is 90. Find the common ratio.

36. How many terms of the GP 3, 3^2, 3^3, ... are needed to give a sum of 120?

37. Find a GP whose 2nd term is 6 and 5th term is 162. Write its first 4 terms.

38. Find the sum of the series 0.6 + 0.66 + 0.666 + ... to n terms.

39. If the pth term of a GP is q and the qth term is p, find its nth term.

40. Three numbers form an increasing GP. If the middle term is doubled, the new numbers are in AP. Find the common ratio.

41. Find the 10th term of the HP 1/3, 1/5, 1/7, ...

42. If the 4th and 9th terms of an HP are 1/10 and 1/20, find the 12th term.

43. Insert two harmonic means between 3 and 12.

44. Find the harmonic mean of 4 and 16.

45. If the AM and GM of two numbers are 10 and 8 respectively, find their HM.

46. The 3rd term of an HP is 8 and the 6th term is 4. Find the 9th term.

47. If the AM and GM of two positive numbers are 13 and 12 respectively, find the numbers.

48. The 5th term of an HP is 1/9 and the 11th term is 1/21. Find the first term of the HP.

49. Find the harmonic mean of 1, 2, 3, and 4.

50. The AM and HM of two numbers are 25 and 9 respectively. Find the numbers.

51. In an HP, the pth term is q and the qth term is p. Find the (pq)th term.

52. Check whether 1/5, 1/8, 1/11, 1/14 are in HP.

53. Find the nth term of the HP 1, 1/3, 1/5, 1/7, ...

54. Find the harmonic mean between 1/3 and 1/9.

55. If 2, x, 8 are in HP, find x.

56. Find the sum to infinity of $1+2x+3x^2+4x^3+...$ ($|x|<1$).

57. Find the sum to infinity of $1+4x+7x^2+10x^3+...$

58. Find the sum to infinity of $2+6x+10x^2+14x^3+...$

59. Find the sum to infinity of $1+3(1/2)+5(1/2)^2+7(1/2)^3+...$

60. Find the sum to infinity of $3+7(1/3)+11(1/3)^2+15(1/3)^3+...$

61. Find the sum to n terms of $1+2x+3x^2+...+nx^{n-1}$.

62. Find the sum to infinity of 1+3/4+5/16+7/64+...

63. Find the sum to infinity of $2+5(1/5)+8(1/5)^2+11(1/5)^3+...$

64. Find the sum to infinity of $1+2(3/4)+3(3/4)^2+4(3/4)^3+...$

65. Find the sum to infinity of $3+6(1/2)+9(1/2)^2+12(1/2)^3+...$

66. Find the sum of the first 5 terms of the series 1+3(2)+5(4)+7(8)+9(16).

67. Find the 4th term of the series whose nth term is $(2n-1)\cdot 3^{n-1}$.

68. Find the sum to infinity of $1+2(1/3)+3(1/3)^2+4(1/3)^3+...$

69. Find the sum to infinity of $5+9(1/2)+13(1/2)^2+17(1/2)^3+\ldots$

70. Find the sum of the first 3 terms of the series $1\cdot3+2\cdot3^2+3\cdot3^3+\ldots$

71. Find the sum to infinity of $4+7(1/4)+10(1/4)^2+13(1/4)^3+\ldots$

72. Find the sum to infinity of $(1/2)+2(1/2)^2+3(1/2)^3+4(1/2)^4+\ldots$

73. Find the sum to infinity of $3(1/3)+6(1/3)^2+9(1/3)^3+\ldots$

74. Find the sum to infinity of $1+5(1/5)+9(1/5)^2+13(1/5)^3+\ldots$

75. Find the sum to infinity of $2+4(1/2)+6(1/2)^2+8(1/2)^3+\ldots$

76. Find the sum of squares of the first 15 natural numbers.

77. Find the sum of cubes of the first 10 natural numbers.

78. Find the sum of squares of the first 20 natural numbers.

79. Find the sum $1^2+3^2+5^2+\ldots+19^2$.

80. Find the sum of cubes of the first 8 even numbers.

81. Find the sum of cubes of the first 5 odd numbers.

82. Evaluate $1^3+2^3+3^3+\ldots+15^3$.

83. Find the sum of squares of the first 7 odd numbers.

84. Find the sum of squares of the first 10 even numbers.

85. If the sum of the first n natural numbers is 210, find n.

86. Find the sum of squares of the first 25 natural numbers.

87. Evaluate the sum of (r^3-r^2) for $r=1$ to 6.

88. Evaluate the sum of $(2r+1)^2$ for $r=1$ to 5.

89. If the sum of squares of the first n natural numbers is 2870, find n.

90. If the sum of cubes of the first n natural numbers is 4356, find n.

91. Evaluate the sum of $r^2(r+1)$ for $r=1$ to 10.

92. Find the sum of squares of the first 12 natural numbers minus the sum of the first 12 natural numbers.

93. Evaluate $1^3-2^3+3^3-4^3+\ldots+9^3-10^3+11^3$.

94. If the sum of squares of the first n natural numbers is 1015, find n.

95. Evaluate the sum of (r^3+3r^2+3r) for $r=1$ to 5.

96. Find the sum of the first 12 terms of the series $1^2+2^2+3^2+\ldots$

97. Find the sum $2^3+4^3+6^3+\ldots+20^3$.

98. Find n if $1^3+2^3+\ldots+n^3=225$.

99. Find the sum of squares of the first n natural numbers in terms of n, and evaluate for $n=30$.

100. Evaluate the sum of r^3 for $r=1$ to 6, plus the sum of r^2 for $r=1$ to 6.

Answer Key

1. 99

2. 62

3. 6, 10, 14

4. 465

5. 16th term

6. 1050

7. $k = -2$

8. 4, 9, 14

9. 670

10. No

11. 1075

12. 40°, 60°, 80°

13. 70336

14. $(pq+1)/2$

15. 12

16. 3, 5, 7

17. 7/16

18. 253

19. $a=-1$, $b=15$, $c=31$

20. $n(3n+1)/2$

21. $a=3$, $r=2$

22. 728

23. 16

24. 8, 16

25. $T_n = 5 \cdot 2^{n-1}$

26. $a = 5$

27. 3, 7, 11

28. 121/81

29. $a=1$, $r=3$

30. 700 m

31. −96

32. 1/3, 1, 3

33. $P^2 = (ab)^n$

34. 3/4

35. $r = \pm 3$

36. 4

37. 2, 6, 18, 54

38. $S_n = (2/3)n - (2/27)(1-(1/10)^n)$

39. $T_n = q^{((n-p)/(p-q))} \cdot p^{((q-n)/(p-q))}$

40. $r = 2+\sqrt{3}$

41. 1/21

42. 1/26

43. 4, 6

44. 32/5

45. 32/5

46. 8/3

47. 18, 8

48. 1

49. 48/25

50. 45, 5

51. 1

52. Yes

53. $1/(2n-1)$

54. 1/6

55. 16/5

56. $1/(1-x)^2$

57. $(1+2x)/(1-x)^2$

58. $2(1+x)/(1-x)^2$

59. 6

60. 15/2

61. $[1-(n+1)x^n+nx^{(n+1)}]/(1-x)^2$

62. 20/9

63. 55/16

64. 16

65. 12

66. 227

67. 189

68. 9/4

69. 18

70. 102

71. 20/3

72. 2

73. 9/4

74. 5/2

75. 8

76. 1240

77. 3025

78. 2870

79. 1330

80. 10368

81. 1225

82. 14400

83. 455

84. 1540

85. 20

86. 5525

87. 350

88. 285

89. 20

90. 11

91. 3410

92. 572

93. 756

94. 14

95. 435

96. 650

97. 24200

98. 5

99. $n(n+1)(2n+1)/6$; 9455

100. 532